

Experimental Study on Feature Selection Methods for Software Fault Detection

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Abstract— Software fault detection is the process of analyzing the software for identifying the errors before it is being deployed to the customer. The classifier is employed to perform the software fault detection. Therefore, the accuracy of the software fault detection highly depends on the classifier which is employed in fault detection. Developing the classifier with irrelevant and redundant features of the error-prone data deteriorates the accuracy in software fault detect. Therefore, the feature selection process is employed to remove the redundant and irrelevant features from the error-prone data to improve the accuracy in the software fault detection. Hence, this paper presents an experimental study on the performance of the feature selection methods namely gain ratio (GR), Info gain (IG), OneR, ReliefF, and symmetric uncertainty (SU) to develop the highly accurate classifier for improving the accuracy in software fault detection.

Keywords—*Feature selection for software fault detection; software fault detection using feature selection; gain ratio-based feature selection; information gain-based feature selection; feature selection for fault detection; study on feature selection; introduction to feature selection.*

I. INTRODUCTION

Software fault detection is a process of identifying the level of fault in the testing phase. Software fault detection plays a major role in improving the quality of the software and to determine the failure of the software so that various losses can be avoided such as improper function of the machine which runs the particular software. The quality of the software highly depends on various software metrics such as accuracy, robustness, error, etc. The defective or failure of the software can lead to many materialistic and physical losses. The outcome of various software metrics of the testing process are considered as the features to develop the classifier for software fault detection. The feature selection is carried out to remove the redundant and irrelevant features from the error-prone data to improve the predictive accuracy in software fault detection. The feature selection is also called as variable or attribute selection [1].

The feature selection algorithms are mainly classified into filter, wrapper, and embedded methods. The filter method uses

the statistical measure to assign a particular score to every feature. The features are given rank based on the score and it is decided based on the priority whether to accept or remove the features from the dataset. Filter methods are mostly univariate and they consider the features either independently or with regard to the features that are dependent. Chi-square test, information gain, and correlation coefficient score are some of the examples of the filter methods. The benefit of using the filter method is that the computational cost is low due to the absence of the classifier. This is highly suitable for high-dimensional data. The wrapper method selects the features based on the search algorithm where various combinations of the features are created, evaluated and compared with others. It is purely based on a predictive model which assigns a score for each feature combination and hence improves the accuracy in the evaluation of the features. This method is computationally inefficient and hence adaptable only for low dimensional data. Embedded methods select the features that contribute to the accuracy of the classification model. They are commonly termed as regularization methods. They select the features which are mostly in the predictive form leading to lower complexity of the model [2] [3].

The rest of the paper is organized as follows. Section 2 reviews the literature, Section 3 discusses the experimental setup and experimental procedure. Section 4 presents the results and discussion, and Section 5 concludes this paper.

II. LITERATURE REVIEW

This section reviews the literatures on the various features selection methods that are carried out by different researchers. Bernhard Schölkopf et al proposed the way of selecting the features by making use of the genetic algorithm and found to yield better results when compared with the existing methods and other cross validation algorithms. Attribute selection is a process of removing irrelevant and redundant features of the machine learning models [4] [5] [6]. Daniel et al applied the algorithm to detect the faulty modules and concluded that the prediction capability is more in the databases with less attributes rather than the original set of the data in the software engineering attributes [7]. Paul M et al described a method for

reducing faults using the method of redundancy in analytical data. It is resolved that the combination of the knowledge-based and analytical scheme can help in the redundancy reduction for easy identification of the faults [8]. Rolf discussed about the need for the fault detection to increase the reliability and safety regarding the usage of the software. The detection process can also be used in the process of monitoring the faults and to estimate various possible earlier detection methods by the usage of some correlation methods [9]. Inseok et al divided the problem into isolation and fault detection. It distinguishes between the uncertainties, noise and other unknown disturbances to predict the fault during the development stage of the system [10]. Martin May et al evaluated the performance of the data using the concept of principal component analysis for the purpose of anomaly detection. The methodology of temporal correlation is adopted to improve the significance of results of the anomaly detection [11].

Dong Xiang et al proposed the usage of many parameters of theoretic measures such as information gain, information cost, and entropy. The utilities of these measures are illustrated by using them in the UNIX system call data [12]. Bernhard Plattner et al developed an approach based on the entropy to determine and report the contents of the entropy which are the main parameters for the traffic in the network. Any changes made in the content of the entropy bring a massive change in the network and provides a good scale of output [13]. N.S.D. Brito et al proposed a novel method for classification and fault detection using the fault transmission data. This method is used to detect the single faults in the neural network classifiers [14]. Taghi M. Khoshgoftaar et al proposed a tool which uses discriminant analysis with the statistical technique for the detection of the program which is fault prone. It is mainly used in the classification of the programs which are of low error rate. This mainly predicts which specific module is error-prone and it can be rectified by the developers during the development of the software [15].

Bernd Kulicke et al presented an approach to obtain a high speed and effective software development. The classification is done based on the various phases of the ground faults. This classification technique can also be employed for systems with very high speed for the protective relay of the systems in the new generation [16]. Andrew H et al discussed the need for the identification and the need for selecting the important features. The selection of significant features makes the problem simple and the solution to be obtained is faster and more accurate. It is proven that the SVM-based feature selection can deliver comparatively higher performance with enhanced results [17]. Gabor-Fisher classifier is mainly applied for the classification of the data based on the facial images. It is suggested that this classifier is more feasible and more accurate when compared with other methods like Eigen faces method [18]. Eric Pottier et al provided a new scheme of solution for the scattering problems on the parameters. The key parameter that is used in the randomness selection is the scattering entropy [19].

Michael described an approach for the texture property characterization using wavelet transform at multiple scales. The comparison for the biorthogonal and orthogonal

performance of the wavelet transform and suggested that discrete wavelet frame can be included in the simple multi-component texture segmentation [20]. Takashi Yamawaki et al suggested that rank correlation gives a good performance for the computational measures. The feature selection is an important process in the selection of relevant features from the extracted set of features [21]. Huan Liu et al discussed that the selection of the features based on relevancy alone is insufficient. So, a new framework is developed using correlation method that concentrates on both relevancy and redundancy analysis to produce effective and efficient results [22].

Vasant Honavar et al identified that the genetic algorithm offers a good approach for the selection of feature subset and to find optimal solution for the problems. The usage of the genetic algorithm produced an efficient result when applied in the neural network automated design for the purpose of knowledge discovery and pattern classification [23]. Karl Pfleger et al performed the selection of the subset features by making use of the supervised induction algorithm to improve the accuracy level using the method of cross validation [24]. Fukunaga et al selected the best subset of features using the method of branch and bound without exhaustive search. This approach is computationally feasible when compared with the dynamic programming and sequential selection which has no guarantee for the optimality of the features [25].

Roberto implemented the concept of subset feature selection using the concept of mutual information which in turn becomes the input for the neural network. The algorithm is done with greedy search among the already selected features [26]. Hak Dinstein et al used the decision rules for the separation of the texture of the images and for classifying the images [27]. A computable set of features is selected to make the classification applications more applicable. Michael Jones et al approached the selection of features using machine learning which is capable of identifying the features from images. This is more applicable for the real time detection of the images without resorting to the differences in the images [28].

Information gain is defined as the difference between the original requirement of the class and the information of the partitioned class. Wolfram Burgard et al made use of the information gain value for estimating the cost of any action. It was used in the vehicle estimation for the process of integrating the approach of exploration, localization and mapping to be applied in the decision theoretic framework. The binary classification method of feature selection makes use of the ReliefF [29]. Feihu et al proposed a method for the selection of the feature subset using the mutual information and reliefF for removing the redundant and irrelevant features, respectively. The combination of ReliefF and the mutual information produce subsets that maintain better classification accuracy compared to using the ReliefF and gain ratio algorithm separately. The derived subset has lesser space complexity and minimized time for computation [30]. The feature selection has also been used for anomaly detection [31]. Waseem Shahzad et al proposed a purely filter based feature selection method

which uses the ranking and the threshold value for the selection of the subset [32].

III. EXPERIMENTAL SETUP AND PROCEDURE

The WEKA data mining tool is used to carry out the experiment. In this experiment, the features are selected from the ten different datasets as tabulated in Table 1 namely Ar1, Ar3, Ar4, Ar5, Ar6, Kc1, Kc2, Kc3, Cm1, and Datatrive (taken from the PROMISE repository that contain various fault detecting features such as cyclomatic complexity, unique operators, decision density, Halstead difficulty, etc.) using the feature selection methods namely Gain ratio attribute evaluation (GR), Information gain attribute evaluation (IG), OneR attribute evaluation (OR), ReliefF attribute evaluation (RF) and Symmetric uncertainty attribute evaluation (SU).

The threshold value $Tv=N/x$ is calculated to determine the number of features to be selected from the dataset where N is the total number of features in the dataset and x is a random number to equally distribute the attributes. Based on the threshold value the features are selected to classify the features using the naive Bayes classifier.

The threshold values are termed as Tv1, Tv2, Tv3, Tv4, and Tv5. The same procedure is carried out for the ten different datasets tabulated in Table 1 using the five different evaluation methods (GR, IG, OR, RF, and SU) and the accuracy results are tabulated.

TABLE I. DETAILS OF THE DATASET

Dataset	Instances	Features	Defect	
			True	False
Ar1	121	29	9	112
Ar3	63	29	8	55
Ar4	107	29	20	87
Ar5	36	29	8	28
Ar6	101	29	15	86
Kc1	2109	21	326	1783
Kc2	522	21	107	415
Kc3	194	39	36	158
Cm1	498	21	49	449
Datatrive	130	8	11	119

IV. RESULTS AND DISCUSSION

The experimental results are tabulated from Table 2 to Table 11 and illustrated from Figure 1 to Figure 10.

TABLE II. ACCURACY OF NAIVE BAYES CLASSIFIER FOR VARIOUS FEATURE SELECTION METHODS WITH SELECTED FEATURES BY TV ON AR1 DATASET

Feature selection method	Tv1	Tv2	Tv3	Tv4	Tv5
GR	85.95	84.29	85.95	85.12	85.95
IG	85.95	84.29	85.95	85.12	85.95
OR	88.42	80.16	81.81	86.77	85.95
RF	85.95	85.12	85.12	84.29	85.95
SU	85.95	84.29	85.95	85.12	85.95

TABLE III. ACCURACY OF NAIVE BAYES CLASSIFIER FOR VARIOUS FEATURE SELECTION METHODS WITH SELECTED FEATURES BY TV ON AR3 DATASET

Feature selection method	Tv1	Tv2	Tv3	Tv4	Tv5
GR	90.47	88.88	88.88	88.88	88.88
IG	90.47	88.88	88.88	88.88	88.88
OR	93.65	88.88	90.47	88.88	88.88
RF	90.47	88.88	88.88	88.88	88.88
SU	90.47	88.88	88.88	88.88	88.88

TABLE IV. ACCURACY OF NAIVE BAYES CLASSIFIER FOR VARIOUS FEATURE SELECTION METHODS WITH SELECTED FEATURES BY TV ON AR4 DATASET

Feature selection method	Tv1	Tv2	Tv3	Tv4	Tv5
GR	85.98	86.91	87.85	86.91	85.98
IG	86.91	86.91	85.98	87.85	85.98
OR	86.91	85.98	86.91	87.85	85.98
RF	81.30	85.04	85.98	81.91	85.98
SU	86.91	85.98	86.91	87.85	85.98

TABLE V. TABLE 5.4 ACCURACY OF NAIVE BAYES CLASSIFIER FOR VARIOUS FEATURE SELECTION METHODS WITH SELECTED FEATURES BY TV ON AR5 DATASET

Feature selection method	Tv1	Tv2	Tv3	Tv4	Tv5
GR	88.88	85.98	87.85	87.85	86.11
IG	88.88	88.88	88.88	88.88	86.11
OR	88.88	88.88	88.88	88.88	86.11
RF	88.88	88.88	88.88	88.88	86.11
SU	88.88	88.88	88.88	88.88	86.11

TABLE VI. ACCURACY OF NAIVE BAYES CLASSIFIER FOR VARIOUS FEATURE SELECTION METHODS WITH SELECTED FEATURES BY TV ON AR6 DATASET

Feature selection method	Tv1	Tv2	Tv3	Tv4	Tv5
GR	88.11	85.14	84.15	86.13	85.14
IG	88.11	85.14	84.15	86.13	85.14
OR	87.12	87.12	88.11	84.15	85.14
RF	84.15	82.17	81.18	85.14	85.14
SU	88.11	85.14	84.15	86.13	85.14

TABLE VII. ACCURACY OF NAIVE BAYES CLASSIFIER FOR VARIOUS FEATURE SELECTION METHODS WITH SELECTED FEATURES BY TV ON KC1 DATASET

Feature Selection method	Tv1	Tv2	Tv3	Tv4	Tv5
GR	83.16	83.12	82.69	82.45	82.55
IG	83.87	83.73	83.07	82.69	82.55
OR	82.97	82.83	82.50	82.78	82.55
RF	80.70	81.31	82.36	82.36	82.55
SU	84.11	83.16	82.93	82.93	82.55

TABLE VIII. ACCURACY OF NAIVE BAYES CLASSIFIER FOR VARIOUS FEATURE SELECTION METHODS WITH SELECTED FEATURES BY TV ON KC2 DATASET

Feature selection method	Tv1	Tv2	Tv3	Tv4	Tv5
GR	83.90	83.33	83.33	83.52	83.71
IG	83.90	83.33	83.14	83.33	83.71
OR	84.67	83.71	84.29	83.90	83.71
RF	82.37	83.71	83.71	83.71	83.71
SU	83.90	83.33	83.33	83.52	83.71

TABLE IX. ACCURACY OF NAIVE BAYES CLASSIFIER FOR VARIOUS FEATURE SELECTION METHODS WITH SELECTED FEATURES BY TV ON KC3 DATASET

Feature selection method	Tv1	Tv2	Tv3	Tv4	Tv5
GR	88.64	87.77	86.24	85.58	84.93
IG	87.11	86.46	86.24	85.37	84.93
OR	84.93	84.93	83.84	84.27	84.93
RF	68.77	77.94	81.44	84.06	84.93
SU	88.42	87.11	86.02	85.80	84.93

TABLE X. ACCURACY OF NAIVE BAYES CLASSIFIER FOR VARIOUS FEATURE SELECTION METHODS WITH SELECTED FEATURES BY TV ON CM1 DATASET

Feature selection method	Tv1	Tv2	Tv3	Tv4	Tv5
GR	87.34	86.54	86.14	85.54	84.93
IG	87.34	86.74	86.14	85.54	85.34
OR	87.95	86.34	84.33	84.33	84.93
RF	85.34	85.54	84.73	85.34	84.93
SU	87.34	86.54	86.14	85.54	84.93

TABLE XI. ACCURACY OF NAIVE BAYES CLASSIFIER FOR VARIOUS FEATURE SELECTION METHODS WITH SELECTED FEATURES BY TV ON DATATRIEVE DATASET

Feature selection method	Tv1	Tv2	Tv3	Tv4
GR	91.53	88.46	87.69	83.07
IG	91.53	88.46	87.69	83.07
OR	87.69	84.61	83.07	83.07
RF	92.30	84.61	84.61	83.07
SU	91.53	88.46	87.69	83.07

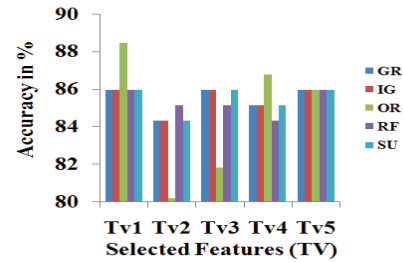


Fig. 1. Accuracy of naive Bayes classifier for various feature selection methods with selected features by Tv on Ar1 dataset

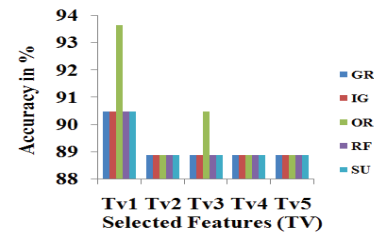


Fig. 2. Accuracy of naive Bayes classifier for various feature selection methods with selected features by Tv on Ar3 dataset

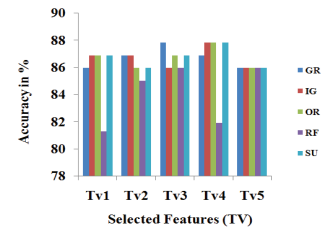


Fig. 3. Accuracy of naive Bayes classifier for various feature selection methods with selected features by Tv on Ar4 dataset

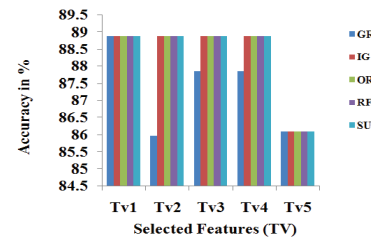


Fig. 4. Accuracy of naive Bayes classifier for various feature selection methods with selected features by Tv on Ar5 dataset

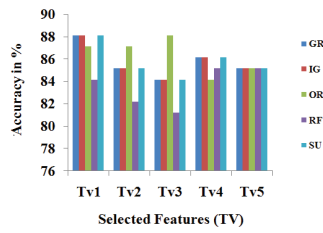


Fig. 5. Accuracy of naive Bayes classifier for various feature selection methods with selected features by Tv on Ar6 dataset

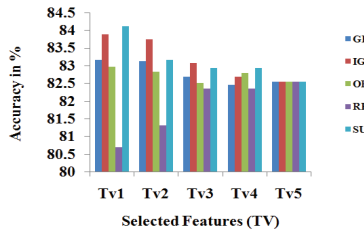


Fig. 6. Accuracy of naive Bayes classifier for various feature selection methods with selected features by Tv on Kc1 dataset

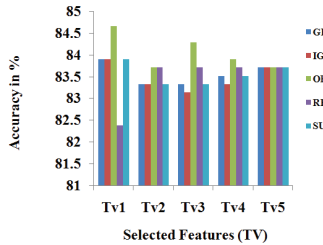


Fig. 7. Accuracy of naive Bayes classifier for various feature selection methods with selected features by Tv on Kc2 dataset

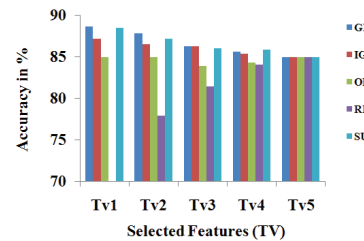


Fig. 8. Accuracy of naive Bayes classifier for various feature selection methods with selected features by Tv on Kc3 dataset

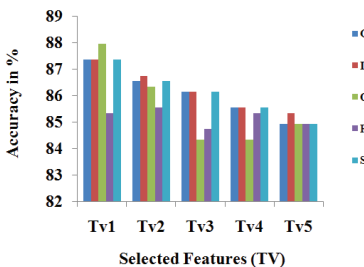


Fig. 9. Accuracy of naive Bayes classifier for various feature selection methods with selected features by Tv on Cm1 dataset

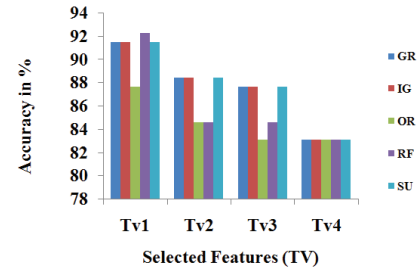


Fig. 10. Accuracy of naive Bayes classifier for various feature selection methods with selected features by Tv on Datatrive dataset

From the experimental results, it is observed that the OR feature evaluation method performs better than other methods. However, there is no significant difference among the feature selection methods in terms of classification accuracy.

V. CONCLUSION

This paper presented an empirical study on the various feature selection methods to improve the performance of the classifier in terms of the classification accuracy for improving the accuracy in software fault detection. In this empirical study, the naïve Bayes classifier is adopted with different feature selection methods namely GR, IG, OR, RF, and SU. The performance of the feature selection methods are tested on ten error-prone software fault detection datasets in terms of classification accuracy of naïve Bayes classifier. From this experimental study it is observed that the OR feature selection method performs better than the other methods compared. Hence, the OR feature selection method is suggested for the prediction of the software fault.

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